PalmSource™ 99

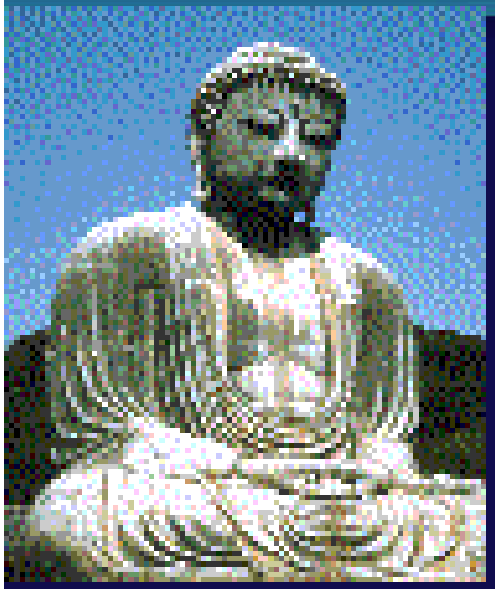
The Zen of Palm™

Product Design for the Soul

Michael Lunsford
Senior Product Manager

3Com
More connected.

The Path to Enlightenment?



- Embrace simplicity
- Focus on what really matters
- Seek the tranquility of the customer
- Use pragmatic innovation
- Don’t run with sharp sticks
- Solve the 12 riddles



Riddle #1

Q. How do you hold the world on your shoulders?



Apple Tried to Hold the PDA World on Its Shoulders

Conventional Wisdom

- Follow the success of the personal computer
- Use cutting edge technology
- Push the envelope
- Cram in as many features as possible
- Respond to customer wish lists
- Add more features, more functionality, more power



Why Didn't the Newton Sell?

PalmSource 99

The Zen of Palm™

- Stand the question on its head
- Don't get sucked into what the competition is doing
- Filter what customers ask for
- Find the real problems
- Create the simplest solution



Riddle #1

Q. How do you hold the world on your shoulders?



Riddle #1

A. Stand on your head.



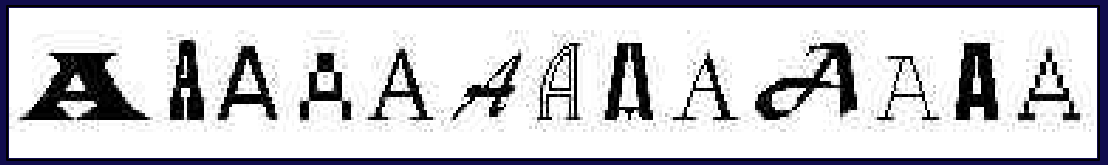
Riddle #2

Q. How can you get a deaf man to hear?



Newton's Ambitious Approach:

Try to get the PDA to understand all the different handwritings in the world.



Palm's Simpler Approach:

Teach the marvelous human brain the simple alphabet of the handheld.



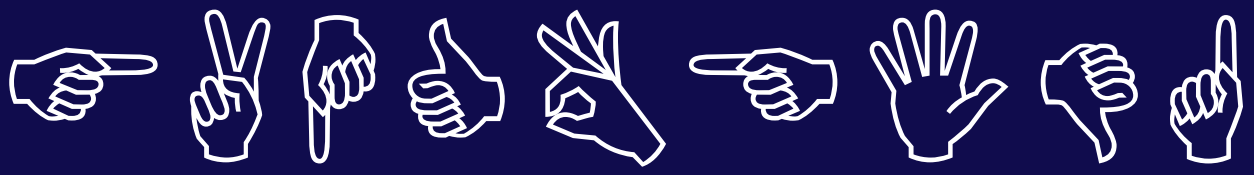
Riddle #2

Q. How can you get a deaf man to hear?



Riddle #2

A. Speak his language.







Riddle #3

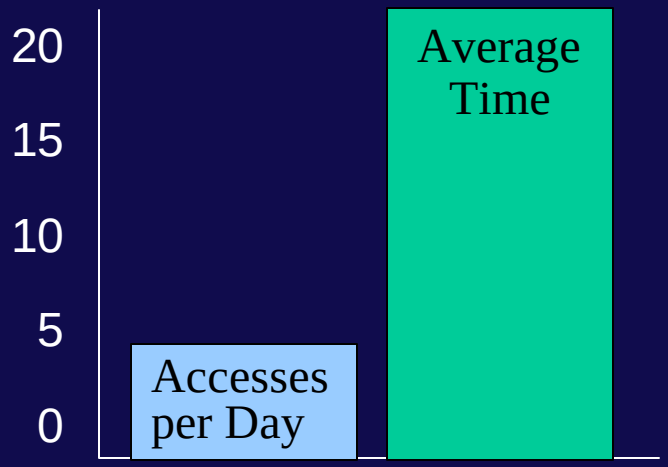
Q. How can a gorilla learn to fly?

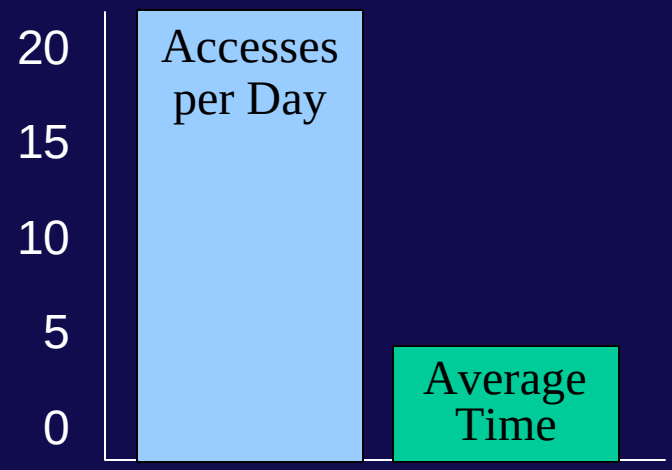




PC	Handheld
Harder to carry	Portable
Bulkier	Smaller
Heavier	Lighter
Long bootup	Faster
More powerful	Simpler







Opposite Usage Pattern

Sales versus Features



PC



Handheld

Riddle #3

Q. How can a gorilla learn to fly?



Riddle #3

A. Only by abandoning the essence of the gorilla.



Riddle #4



Q. Who is Microsoft's biggest enemy?



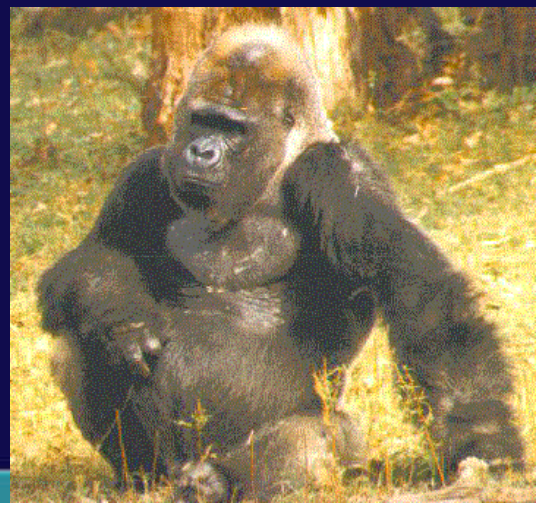
Microsoft's Thinking...

PalmSource 99

More features are always better!

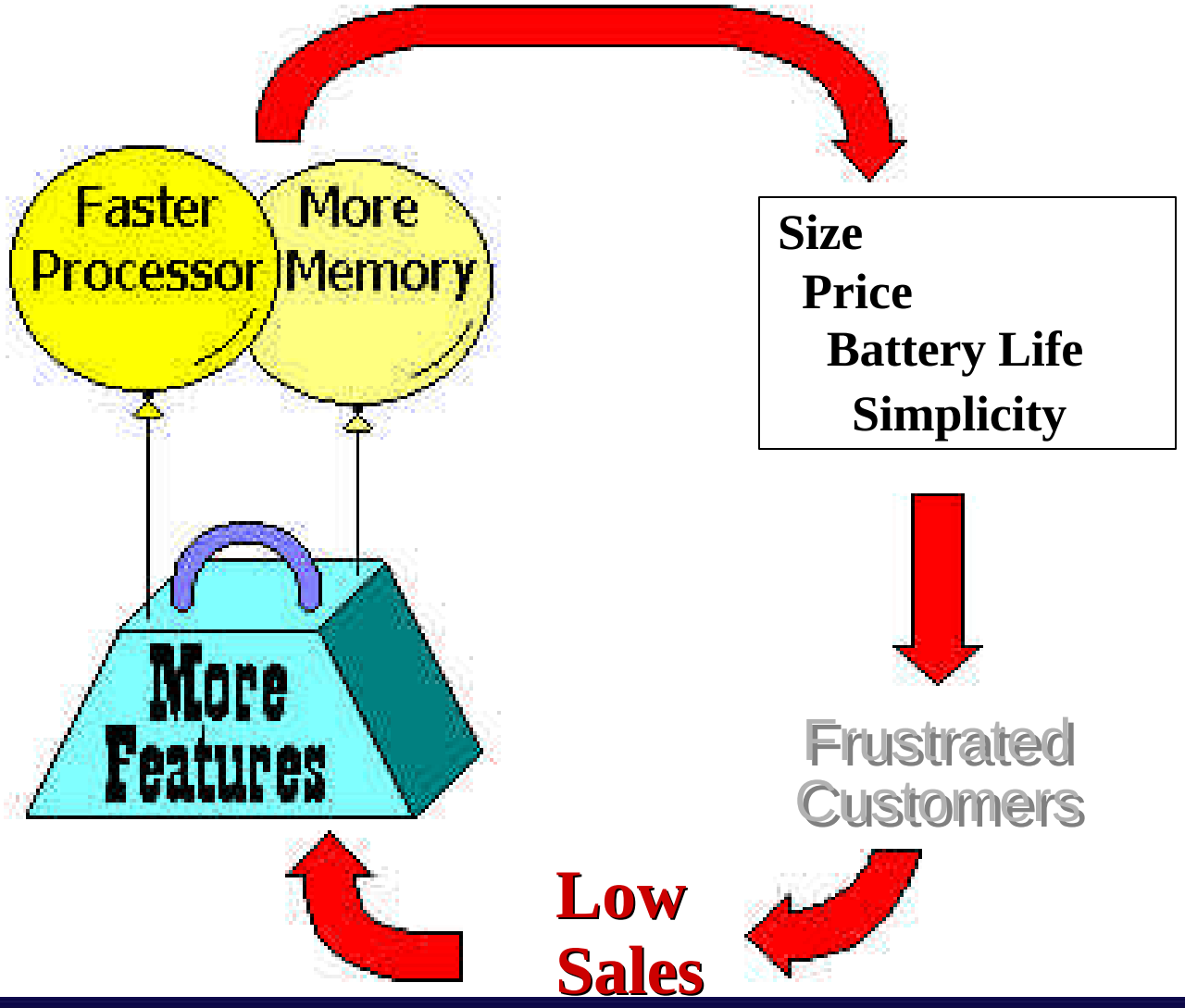
"You want people to have more processing power, a bigger hard drive, more DRAM, a better user experience..."

Craig Barrett
CEO, Intel



The Death Spiral

More features weigh you down!



Riddle #4



Q. Who is Microsoft's biggest enemy?



Riddle #4

A. Microsoft.

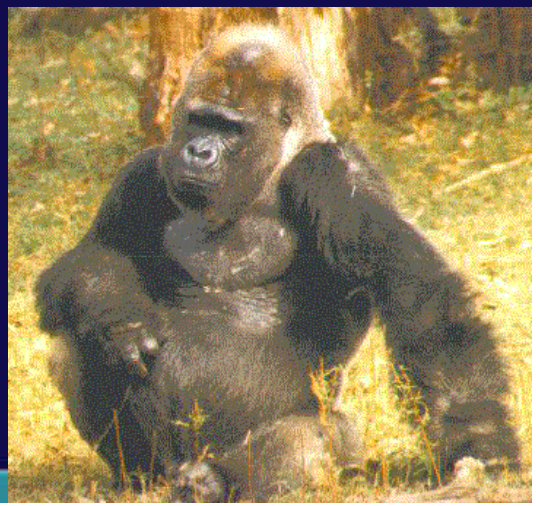


What to Watch Out For



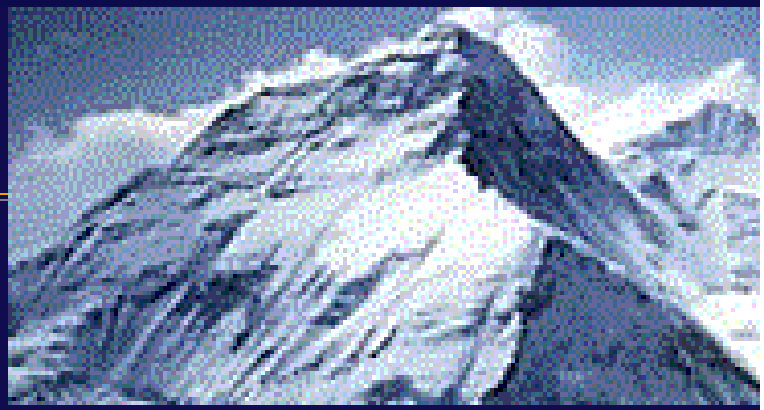
Although their basic instincts lead to failure in our world…

- Microsoft will keep trying to win
- If you're flying ahead, keep your lead
- The danger is if *we* copy *them*



Riddle #5

Q. How do you put a mountain in a teacup?





Nirvana: the Sweet Spot

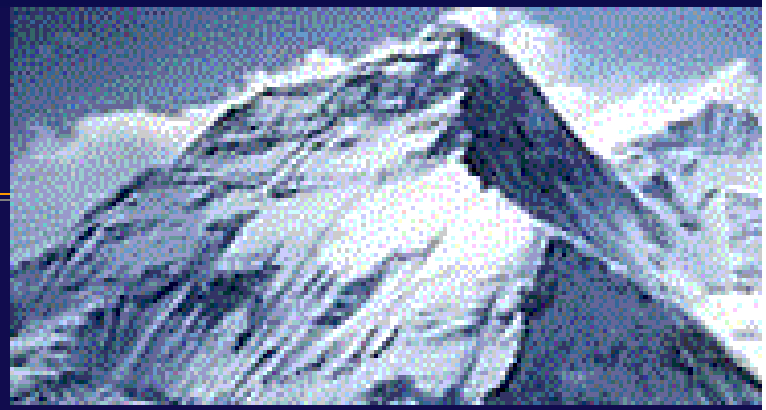
- Size
- Price
- Battery Life
- Ease of Use



- Speed
- Efficient Memory
- Connectivity
- Platform
- Screen Display
- Style

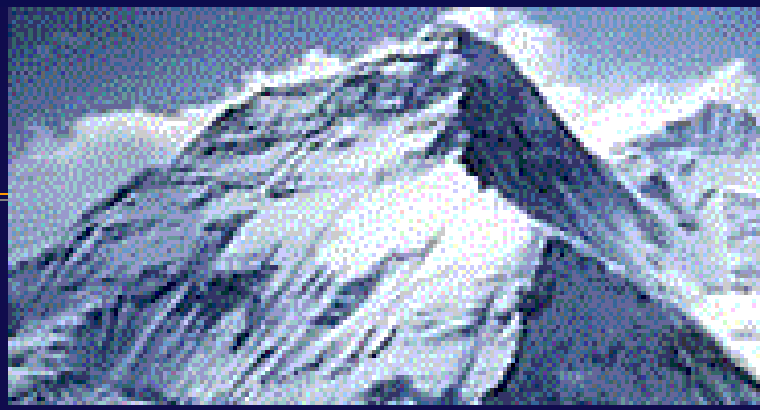
Riddle #5

Q. How do you put a mountain in a teacup?





Riddle #5



A. Dig for a few diamonds and put *them* in the cup.



(Who needs all that dirt and rocks?)

Riddle #6

Q. What can you do to quiet the whining of a spoiled kid?



What Are Customers Screaming For?

PalmSource 99

WinCE

- Windows everywhere
- All the complexity of Windows in the palm of your hand
- The PC model in a handheld
- Lots of built-in features
  - Voice recorder
  - MP3 player
  - Vibrating alarm
  - …more cheese!



What Do Customers Really Need?

PalmSource 99

Palm™ connected organizer

- An easy-to-carry, easy-to-use handheld
- Simplicity and speed
- Great battery life
- “Technology that doesn’t suck”

Rule of Thumb

Don’t give customers *everything* they ask for (it isn’t good for them)

How to Gauge Success

PalmSource 99

Benchmarks

- The Phone Test
- No Wait cursor



Theme

- "Watch" usage test



Strategy

- Minimize taps
- Make it intuitive (or at least easy to remember)

Riddle #6

Q. What can you do to quiet the whining of a spoiled kid?



Riddle #6

A. Wear earplugs.



Riddle #7

Q. How can you give a baby the world?



Zen of Customer Consciousness

PalmSource 99

- Filter what customers ask for
- Study what customers *do*
- Figure out the customer's *real* needs
- Determine the simplest solution

Wants	*Real Needs*
More memory	⇨ Efficient use of memory
Keyboard	⇨ Simple, easy-to-use data entry
Spreadsheet	⇨ A place to hold numbers & lists
Faster Processor	⇨ Instant access to data

Customers Want Speed

PalmSource 99

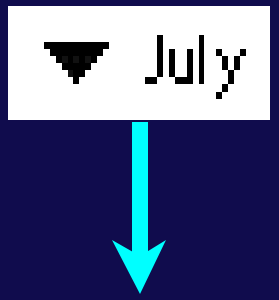
- Fast HotSync® technology
- Application buttons
- Finger navigation
- Edit-in-place
- Uncomplicated menu choices

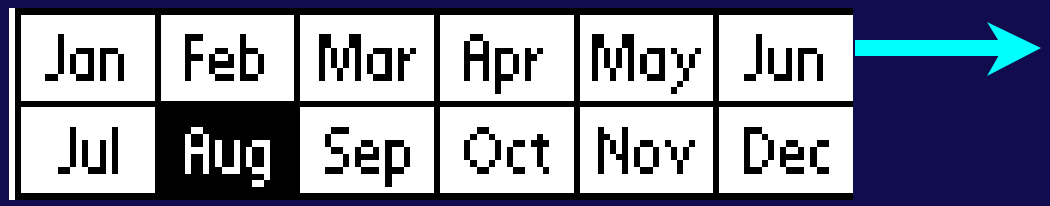


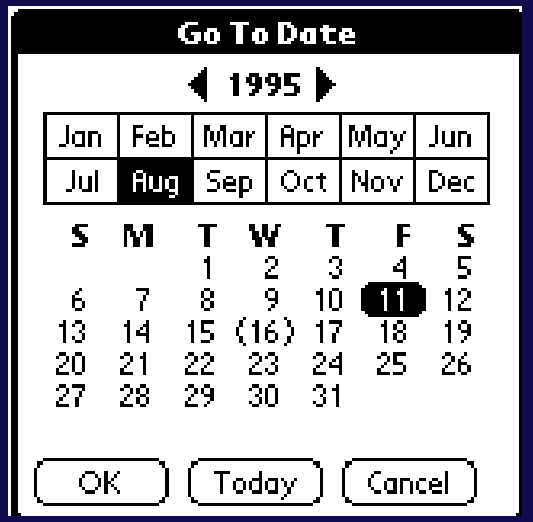
Better Fast than Pretty

Go to Date dialog: month selection options

Pick list







Push buttons

Features vs. Solutions

Don’t ask:

“How do I add more features?”



Instead, ask:

“What does the user need to do?”



Riddle #7

Q. How can you give a baby the world?



Riddle #7

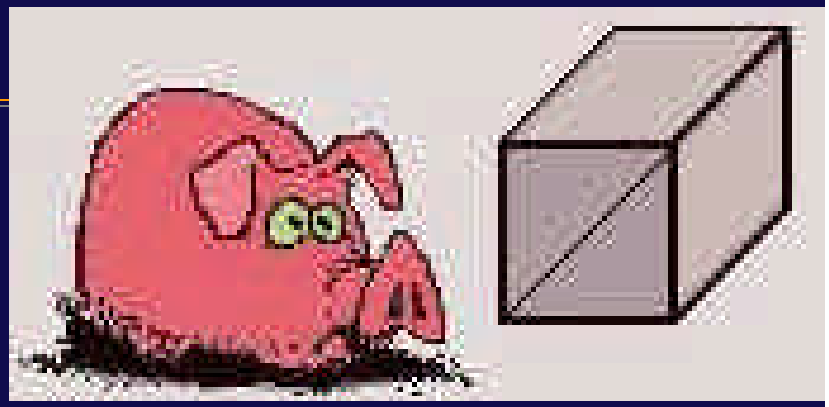
A. Find out the real need
and give the baby that.



(When a baby wants to be changed,
that is the world for the baby.)

Riddle #8

Q. How do you fit a round pig into a square hole?



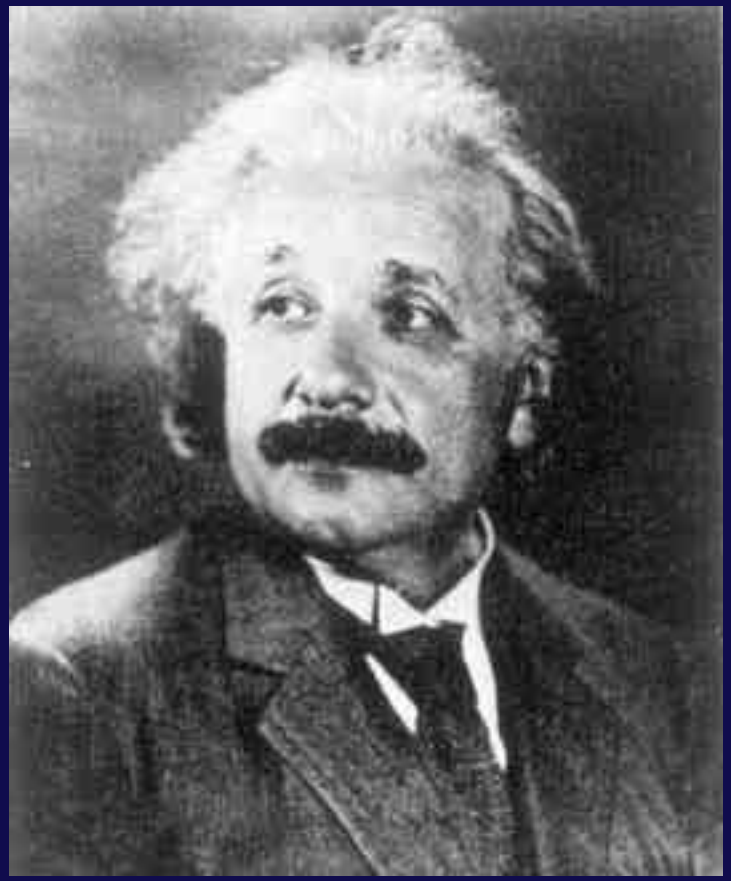
Palm Computing's UI Goals

- Fewer steps to get to your data
- One button access
- Fewer stylus taps
- No indecipherable icons
- No Start menu
- Simple, clean interface





"Make things as simple as possible,
but not simpler."



- Albert Einstein

$E = MC^2$

Simplicity Wins

- Adding lots of extra functionality has tradeoffs in handheld products
- Define a product like packing a backpack
- Most customers ignore most features
- Simplicity is power



Power = Getting the Job Done

- Slower processor → better performance
- Smaller batteries → longer battery life
- No bells and whistles → coolest product
- Less functionality → more useful

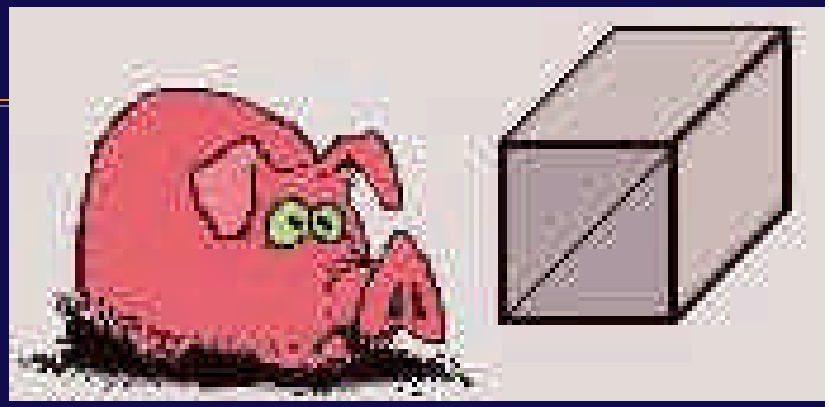






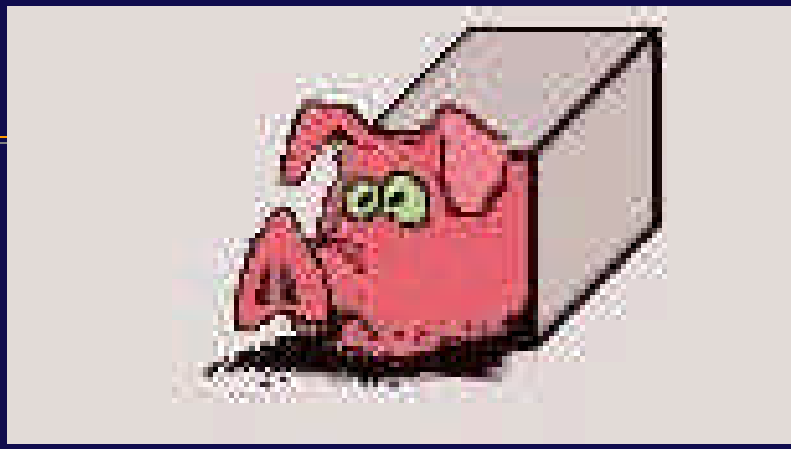
Riddle #8

Q. How do you fit a round pig into a square hole?



Riddle #8

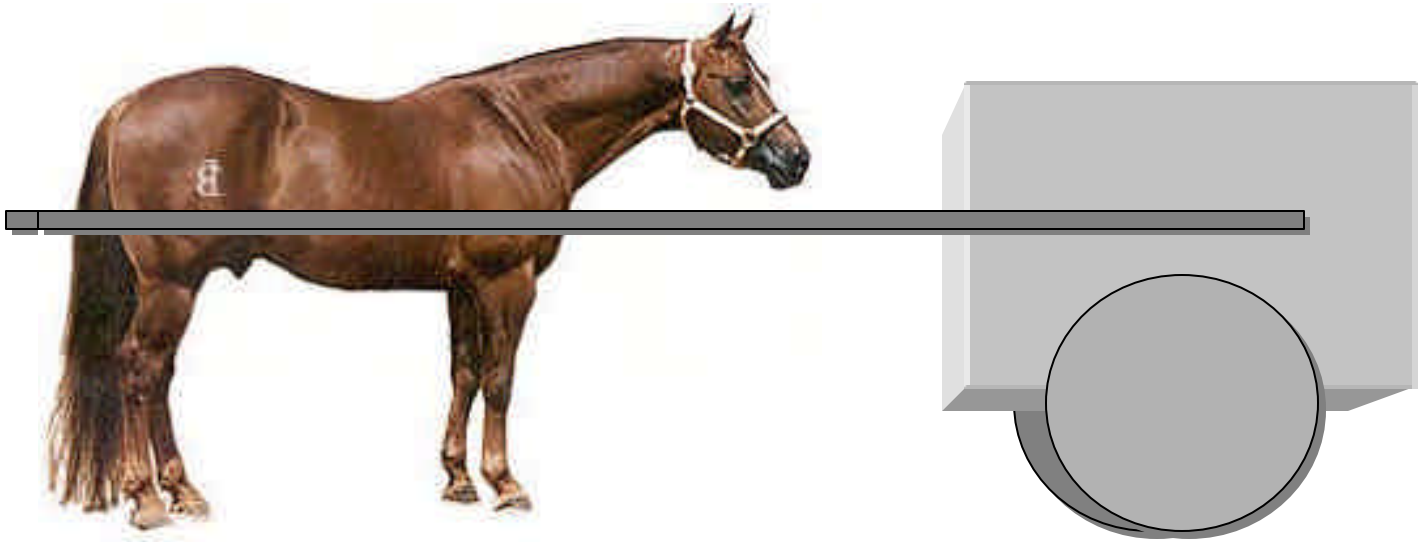
A. Put the pig on a diet.



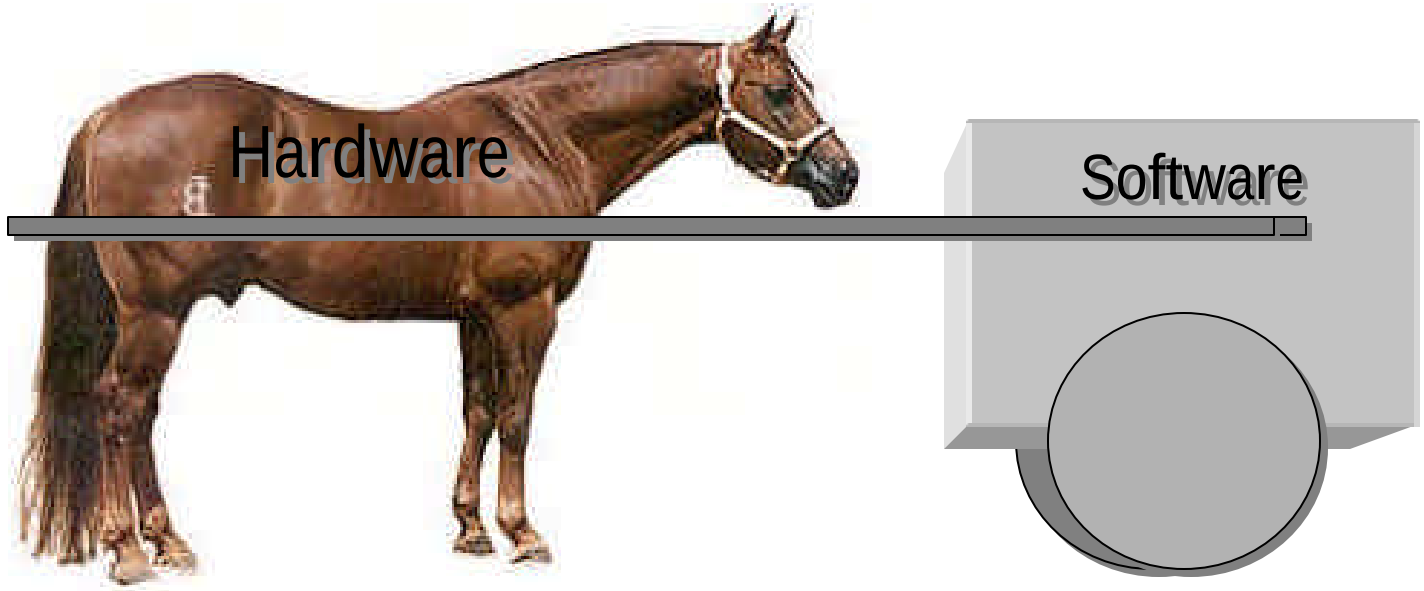
(Less is more.)

Riddle #9

Q. How can you avoid getting the cart before the horse?



In the PC world, software has always pulled the hardware along.



But this doesn't work if you're trying to give birth to a new category

The Electric Car

PalmSource 99

Benefits

- Ecologically friendly, fights pollution
- Runs on renewable energy
- Has the convenience of home charging

Disadvantages

- Tiny cars
- More expensive
- Not enough power or distance
- Reduced mobility
- Limited flexibility

Size

Price

Battery life

Ease-of-use

So nobody buys—except early adopters

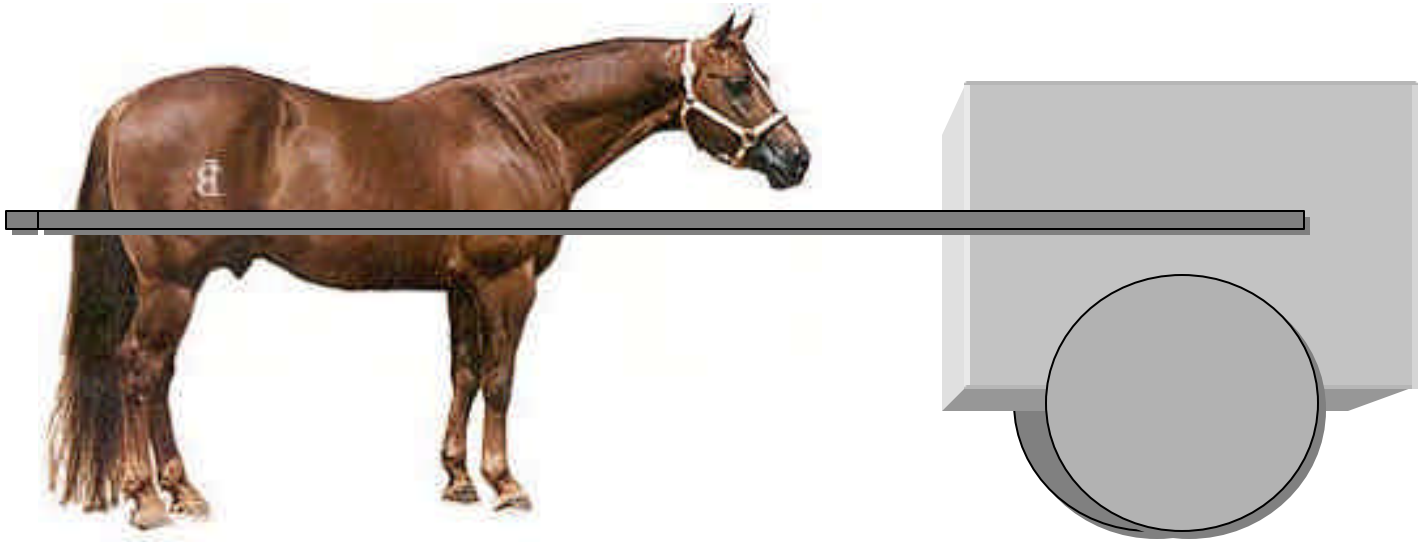
To Stay in the Sweet Spot:

PalmSource 99

Don’t go so far out in front that you leave vital customer needs behind

Riddle #9

Q. How can you avoid getting the cart before the horse?



Riddle #9

A. Put the horse inside the cart.



Riddle #10

Q. How can you age wine faster?



Pragmatic Innovation

PalmSource 99

- Don’t use bleeding-edge technology just because it’s cool
- First look for innovative uses of trailing-edge, proven technologies that will delight the customer
- (The Dragonball processor is tried-and-true)
- Don’t ask, “Where can we use this technology?”
- But rather, “What are the customers’ needs?”

What Are the Customers' Wireless Needs?

PalmSource 99

- The Holy Grail of handheld computing:
  - Ubiquitous access to timely, useful information
- What's prevented success in the past?
  - Too difficult: Customer is the system integrator
  - Too big (WinCE +Wireless Modem = 2.5 Palm III™connected organizer)
  - Too expensive ($800–$1500 + monthly fees >$30)
  - Too slow (www.disney.com takes 3 minutes!)
  - Poor battery life

Palm VII™ Organizer Objectives

PalmSource 99

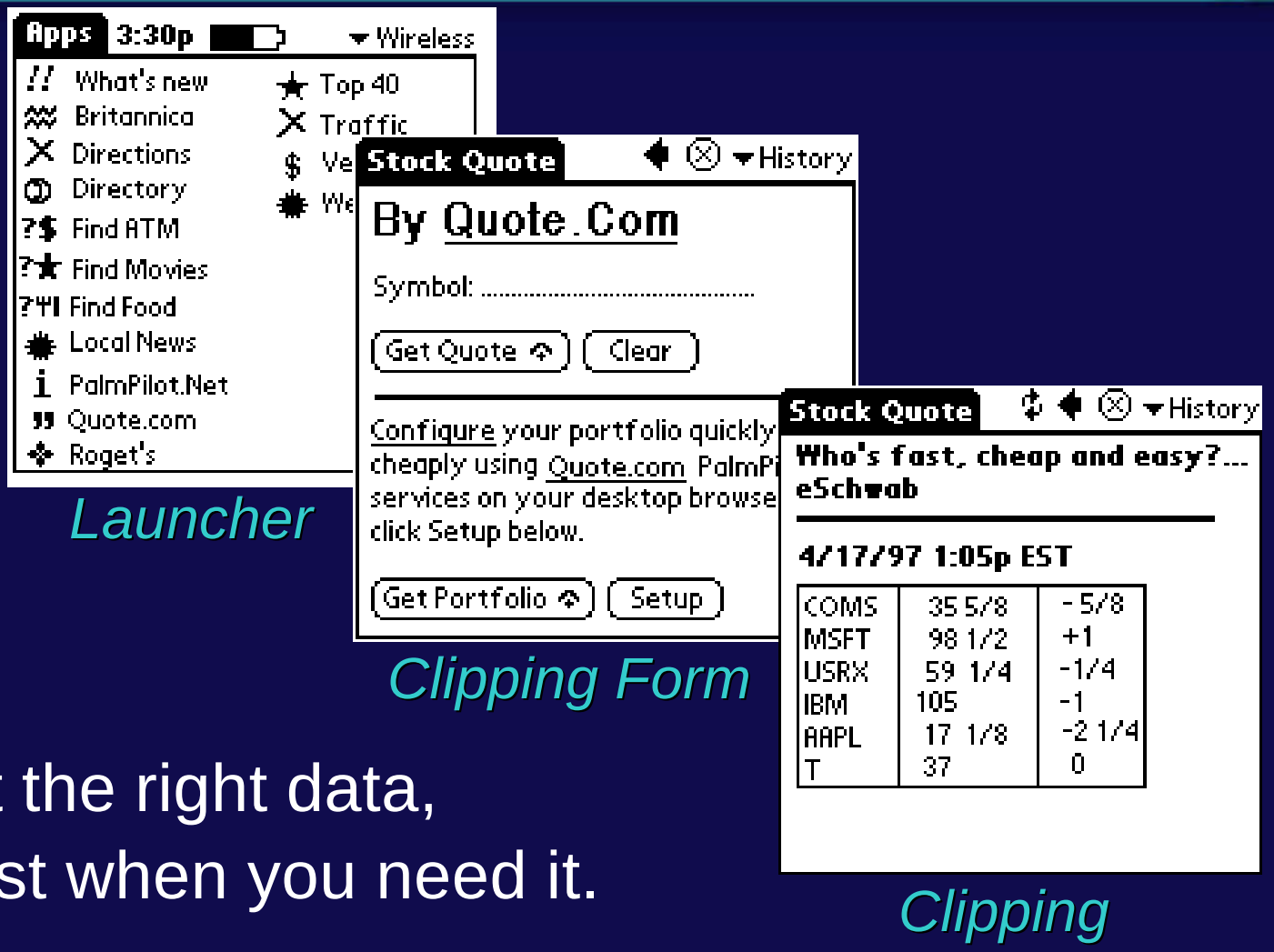
- First and foremost a Palm™ connected organizer
  - Palm OS® software functionality, Palm form factor
- Instant access to useful, timely content
  - Web Clipping
- Short, two-way messaging
- “Zero configuration” out-of-box experience
- Battery life: measured in weeks, not hours
- Decent price: $499 + $9.95/mo. service fee

Wireless Web Browsing on a Handheld?

- Poor performance due to relatively low bandwidth and high latency
- Cost at every transaction in both time and money
- Content presented in 640x480 format
- Huge battery requirements



Web Clipping



Just the right data,
just when you need it.

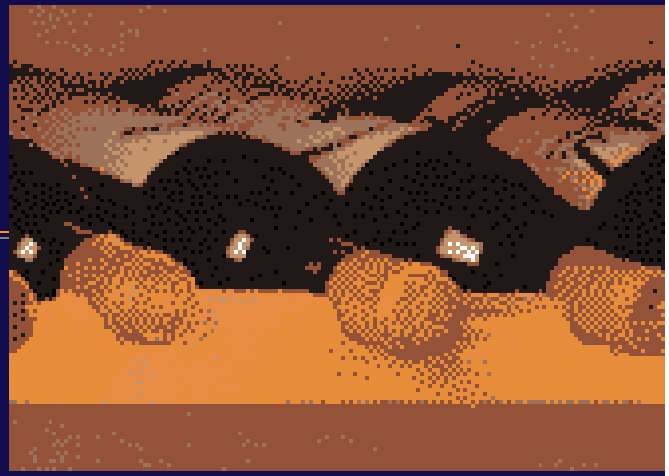
Riddle #10

Q. How can you age wine faster?



Riddle #10

A. Start with older wine.



Riddle #11



Q. How can you roller skate in a buffalo herd?

Palm V™ Organizer

- Lithium-ion batteries
- Advanced LCD technology
- State-of-the-art sealing methods



Riddle #11



Q. How can you roller skate in a buffalo herd?

Riddle #11

A. You can't.

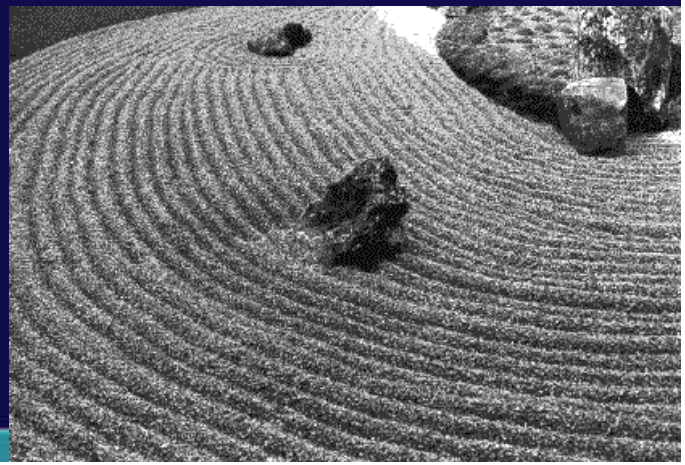


(But with a buffalo stampede behind you, maybe it's time to invest in a motorcycle.)

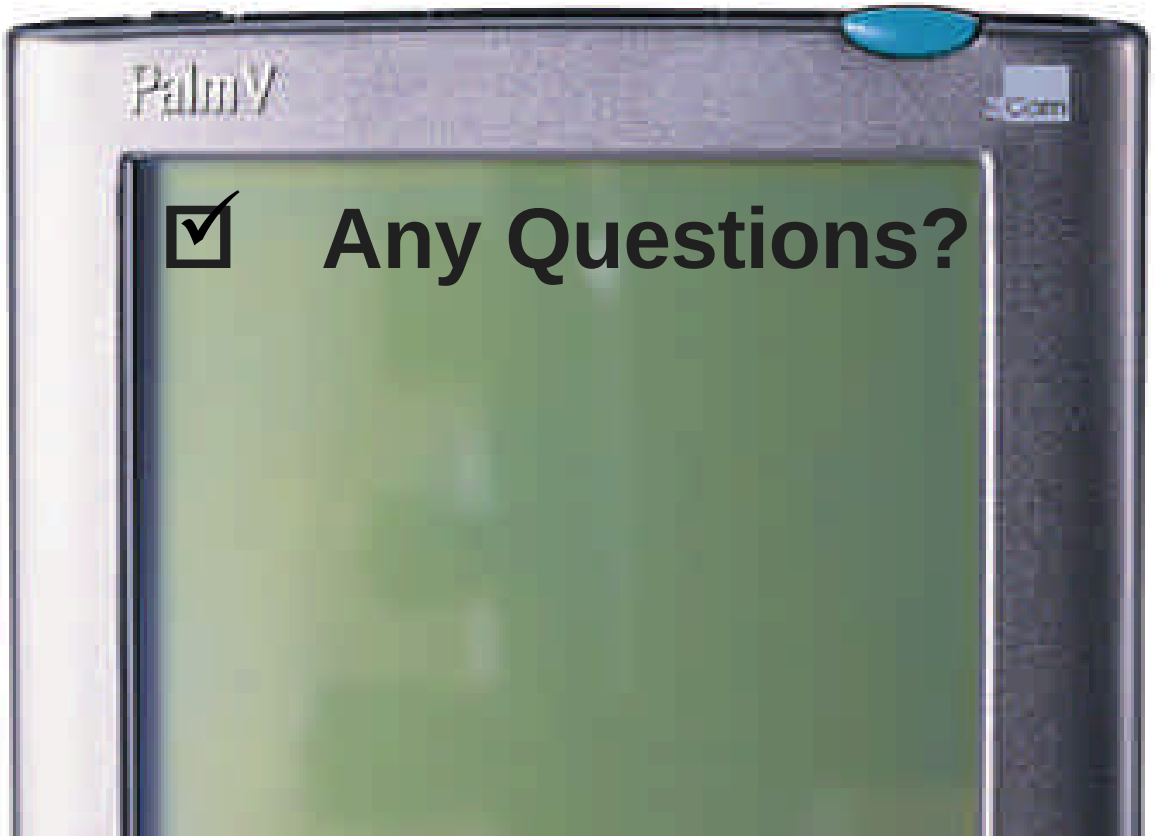
Summary: The Zen Approach

PalmSource 99

- Abandon your PC instincts
- Strive for simplicity
- Forego pleasures of material features
- Focus on the customer's inner tranquility
- Punt the rest



Riddle #12




PalmSource™ 99
3Com
More connected.